CS 201 Graph Algorithms

Week 9: Graph Traversal & Shortest Paths

1. Graph Representations
   - Adjacency list vs adjacency matrix
   - Directed vs undirected graphs
2. Breadth-First Search (BFS)
   - Uses a queue, explores level by level
   - Finds shortest path in unweighted graphs
3. Depth-First Search (DFS)
   - Uses a stack (or recursion)
   - Useful for cycle detection, topological sort

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4. Dijkstra's Algorithm
   - Greedy approach, uses a priority queue
   - Works only with non-negative edge weights
5. Bellman-Ford Algorithm
   - Handles negative edge weights
   - $O(V*E)$ time complexity
6. Complexity Comparison
   - BFS/DFS: $O(V + E)$
   - Dijkstra: $O((V + E) \log V)$ with a heap
   - Bellman-Ford: $O(V * E)$